Concurrency:Multithreading

Concurrency – bu bir nechta vazifani bir vaqtda bajarish imkoniyati. Python’da bu I/O operatsiyalari, ko‘p oqimli (multithreading), ko‘p jarayonli (multiprocessing) yoki asinxron dasturlash (asyncio) orqali amalga oshiriladi. Concurrency haqiqiy parallel ishlashni emas, balki vazifalar o‘rtasida vaqtni samarali taqsimlashni anglatadi.

Multithreading haqida

Multithreading – bu bitta jarayon ichida bir nechta oqimni (thread) bir vaqtda bajarishga imkon beruvchi dasturlash texnikasi. Har bir oqim o‘z vazifasini bajaradi va ular xotirani bo‘lishib ishlaydi. Multithreading asosan **I/O intensiv** vazifalarda ishlatiladi.

```
# Thread yaratamiz
t1 = threading.Thread(target=print_hello)
t2 = threading.Thread(target=print_world)
# Threadlarni ishga tushiramiz
t1.start()
t2.start()
# Barcha thread tugaguncha kutamiz
t1.join()
t2.join()
print("Barchasi tugadi!")
```

```
import threading
import time
def kutish():
    print("Fayl o‘qilmoqda...")
    time.sleep(5)
    print("O‘qish tugadi.")
def salom_berish():
    for _ in range(5):
        print("Assalomu alaykum!")
        time.sleep(1)
t1 = threading.Thread(target=kutish)
t2 = threading.Thread(target=salom_berish)
t1.start()
t2.start()
t1.join()
t2.join()
```

```
def print_user(name):
    for i in range(3):
        print(f"{name} ishlayapti")
        time.sleep(1)
users = ['Ali', 'Vali', 'Soli']
threads = []
for user in users:
    t = threading.Thread(target=print_user, args=(user,))
    threads.append(t)
    t.start()
# Hamma threadlar tugashini kutamiz
for t in threads:
    t.join()
print("Hammasi tugadi")
```

Multithreading’ning afzalliklari

- **Tezlik**: Bir nechta vazifalarni bir vaqtda bajarib vaqtni tejaydi.
- **Resurslarni tejash**: Xotira almashish tufayli resurslardan samarali foydalanadi.
- **Interaktivlik**: Foydalanuvchi bilan o‘zaro ta’sirni yaxshilaydi, masalan, dastur sekinlashmaydi.